A proof of the Kauers–Koutschan conjecture for OEIS A181280

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Abstract

We prove Conjecture 20 of Kauers and Koutschan (arXiv:2303.02793): the number $a(n)$ of $4 \times n$ binary matrices M with rows in strictly increasing order and rows of $MM^{\mathsf{T}} \pmod 2$ in strictly decreasing order equals an explicit exponential polynomial, for all $n \geq 0$. The proof constructs a finite-state column automaton for the count, certifies an order-16 linear recurrence for a via an exact linear-algebra identity for the transfer operator, and closes the gap to the conjectured formula with a finite check.

1 The sequence and the conjecture

For a $4 \times n$ matrix M over $\mathbb{F}_2$, read each row as a binary number (first column most significant). OEIS A181280 (R. H. Hardin, 2010) defines $a(n)$ as the number of such matrices whose four rows are strictly increasing while the four rows of the Gram matrix $G = MM^{\mathsf{T}} \bmod 2$ (each row read as a 4-bit number, first column of G most significant) are strictly decreasing. The convention is fixed by the worked example in the OEIS entry, and our automaton below reproduces all 27 terms of the OEIS b-file.

Theorem 1 (= Kauers–Koutschan Conjecture 20)**.** *For all $n \geq 0$,*

$$\begin{aligned} a(n) \;&= \tfrac{1}{3}\, 2^{2n-11} \left(6n^2 - 219n + 820\right) - \tfrac{1}{9}\, 2^{n-5}(3n+32) - \tfrac{113}{3}\, (-1)^n\, 2^{3n-14} \\ &\quad + 2^{4n-9} - \tfrac{1}{3}\, (-1)^n\, 2^{2n-11}(13n-164) + \tfrac{1}{9}\, 2^{3n-14}(288n - 3473) \;=:\; F(n). \end{aligned}$$

(In particular $F(0) = F(1) = F(2) = F(3) = 0$; the OEIS entry states the formula as a conjecture without specifying its range of validity, and it in fact holds from $n = 0$ on.)

2 A column automaton

Process M column by column; a column is a vector $c \in \mathbb{F}_2^4$. Define the state after reading a prefix of columns as the pair (B, G) where:

- B is the ordered partition of the row indices $\{1,2,3,4\}$ into maximal blocks of *consecutive* rows whose prefixes are equal so far;

*Produced in a two-agent collaboration set up by Jonathan Bostock, 2026-07-22. Adversarial review by OpenAI Codex pending at compile time; see the repository's `STATE.md`. Scripts: `work/claude/a181280_dp.py`, `work/claude/a181280_certify.py`.

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- $G \in \mathbb{F}_2^{4\times 4}$ is the symmetric matrix $\sum_c cc^{\mathsf{T}}$ accumulated over the columns read so far (10 bits).

Lemma 2. *The blocks of B are always intervals of consecutive indices, and a prefix of columns is extendable to a matrix with strictly increasing rows if and only if every column so far splits each then-current block into (rows with bit 0, followed by rows with bit 1), lower indices receiving 0.*

Proof. Two rows compare by their bits at the first column where their prefixes differ, 0 meaning smaller (most-significant-first reading). Suppose inductively the tied blocks are intervals (initially the single block $\{1, 2, 3, 4\}$). When a column assigns bits within a tied block, every row receiving 0 becomes smaller than every row receiving 1, and this decides all pairs between the two parts permanently, with no further constraints between them. Strict increase therefore requires the 0-part to be exactly the lower-indexed rows; both parts are again intervals. Pairs in distinct blocks were decided correctly at an earlier column; pairs within a block remain tied. Conversely, if every split respects the index order, any completion that eventually separates all rows in this manner yields strictly increasing rows. □

There are 8 interval partitions (compositions of 4) and 2^{10} values of G, hence 8192 states. The transition on a column c rejects (kills the path) if c violates Lemma 2, and otherwise refines B and replaces G by $G \oplus cc^{\mathsf{T}}$. Acceptance after n columns: B is the partition into singletons (all rows separated — in particular, no two rows equal), and the rows of G, as 4-bit numbers, strictly decrease. Since $G_{ij} = \langle r_i, r_j\rangle \bmod 2$ is determined by the accumulated outer products, the automaton computes exactly the defining conditions, and

$$a(n) = u_0^{\mathsf{T}} T^n w,$$

where T is the 8192×8192 transition operator (nonnegative integer matrix), u_0 the initial state indicator, and w the acceptance indicator. As validation, this reproduces all 27 terms of the OEIS b-file.

3 A certified recurrence

Proposition 3. *Let $v_k = u_0^{\mathsf{T}} T^k$ (a row vector). The vectors $v_0, \dots, v_{15}$ are linearly independent and $v_{16} \in \operatorname{span}_{\mathbb{Q}}(v_0, \dots, v_{15})$; writing $v_{16} = \sum_{j=0}^{15} c_j v_j$ and $q(t) = t^{16} - \sum_j c_j t^j$, we have the exact factorization*

$$q(t) \;=\; t^4\,(t-16)(t-8)^2(t-4)^3(t-2)^2(t+2)(t+4)^2(t+8).$$

Consequently $\sum_{j=0}^{16} q_j\, a(n+j) = 0$ for all $n \geq 0$.

Proof. The dependence is found and verified coordinate-by-coordinate in exact rational arithmetic on the full 8192-dimensional vectors (only 1242 states are reachable, which speeds this up but is not needed for correctness). Applying T^n on the right of the identity $\sum_j q_j v_j = 0$ gives $\sum_j q_j v_{n+j} = 0$ for every $n \geq 0$, and pairing with w gives the recurrence for $a(n) = v_n w$. □

4 From the recurrence to the closed form

Let

$$p(t) \;=\; (t-16)(t+8)(t-8)^2(t-4)^3(t+4)^2(t-2)^2, \qquad \deg p = 11,$$

the characteristic polynomial of the shift-invariant space of exponential polynomials containing F (bases 16^n; $(-8)^n$; $8^n, n8^n$; $4^n, n4^n, n^2 4^n$; $(-4)^n, n(-4)^n$; $2^n, n2^n$).

Lemma 4. *$\sum_{i=0}^{11} p_i F(n+i) = 0$ for all n.*

Proof. Symbolic verification (sympy): substituting F and simplifying the exact rational-exponential expression yields identically 0. (Equivalently, this is the standard fact that $(S-\lambda)^{d+1}$ annihilates $n^d \lambda^n$; the symbolic check removes any bookkeeping doubt.) □

Proof of Theorem 1. Write $q(t) = t^4\, \tilde{q}(t)$ with $\tilde{q}(0) \neq 0$ (Proposition 3), and let $r = \operatorname{lcm}(\tilde{q}, p)$; here $\tilde{q} \mid p \cdot (t+2)$ and one computes $\deg r = 12$ and $r(0) \neq 0$. By Proposition 3, the shifted sequence $A(n) := a(n+4)$ satisfies the $\tilde{q}$-recurrence for all $n \geq 0$, hence also the r-recurrence; by Lemma 4, $n \mapsto F(n+4)$ satisfies the p-recurrence for all n, hence also r. Two sequences satisfying the same order-12 linear recurrence on all windows $n \geq 0$ and agreeing at 12 consecutive indices are equal; the values $a(n) = F(n)$ for $n = 4, \ldots, 15$ are verified exactly. Hence $a(n) = F(n)$ for all $n \geq 4$. Finally $a(n) = 0 = F(n)$ for $n = 0, 1, 2, 3$ is checked directly (a: no $4 \times n$ matrix with $n \leq 3$ has four distinct rows; F: exact evaluation). □

Remark 5 (What is computer-verified). The b-file agreement (27 terms); the Krylov dependence and its coordinate-wise verification (exact rational arithmetic); the factorization of q; the symbolic annihilation $p(S)F \equiv 0$; $\deg r = 12$, $r(0) \neq 0$; and the 16 value agreements. All computations are integer or rational exact; total runtime a few seconds. Everything is reproducible from `a181280_certify.py`.